

Definition 6.46 (orthogonal complement): Let U a subset of V , then the orthogonal complement of U is denoted by $U^\perp$, a set of all vector in V that orthogonal to U :

$$U^\perp = \{v \in V \mid \forall u \in U, \langle u, v \rangle = 0\}$$

Theorem 6.47 (Properties of orthogonal complement):

- a) If U a subset of V , then $U^\perp$ is a subspace of V
- b) $\{0\}^\perp = V$
- c) $V^\perp = \{0\}$
- d) If U a subset of V , then $U \cap U^\perp \subseteq \{0\}$
- e) If G, H are subset of V and $G \subseteq H$, then $H^\perp \subseteq G^\perp$

Proof.

- 1. $0 \in U^\perp$ cause 0 is orthogonal to any vector.
For any $v, w \in U^\perp$ and $u \in U$, $\langle u, v + w \rangle = \langle u, v \rangle + \langle u, w \rangle = 0$.
For any $\lambda \in F$, $w \in U^\perp$ and $u \in U$, $\langle u, \lambda w \rangle = \bar{\lambda} \langle u, w \rangle = 0$.
- 2. Trivial
- 3. For any non-zero $v \in V^\perp$, we have $\langle v, v \rangle = 0$, thus $v = 0$.
- 4. For any $u \in U \cap U^\perp$, we have $\langle u, u \rangle = 0$, thus $u = 0$.
- 5. For any $v \in H^\perp$, then for any $g \in G$, we have $\langle v, g \rangle = 0$ since $g \in G \subseteq H$. Thus $H^\perp \subseteq G^\perp$. ■

Theorem 6.49: Let U a finite subspace of V , then

$$V = U \oplus U^\perp$$

Proof. We will show $V = U + U^\perp$ and $U \cap U^\perp = \{0\}$.

- For any $v \in V$, since U is a finite subspace of V , let $v_0, \dots, v_{m-1}$ a basis of U ,

then applying Gram-Schmidt Procedure on $v_0, \dots, v_{m-1}, v$, we have $e_0, \dots, e_{m-1}, e$, where e is a linear combination of $v_0, \dots, v_{m-1}$ and v . Then we know $e_0, \dots, e_{m-1} \in U$, and e is orthogonal to all $e_0, \dots, e_{m-1}$, thus $e \in U^\perp$. Since e is a linear combination of $v_0, \dots, v_{m-1}, v$, then we can write v as a linear combination of $v_0, \dots, v_{m-1}, e$ where $v_0, \dots, v_{m-1} \in U$ and $e \in U^\perp$, thus $V = U + U^\perp$.

- We know $U \cap U^\perp \subseteq \{0\}$ and both U and $U^\perp$ are subspace of V , then $0 \in U$ and $0 \in U^\perp$, thus $U \cap U^\perp = \{0\}$.

For part 1, let $u = \langle v, e_0 \rangle e_0 + \dots + \langle v, e_{m-1} \rangle e_{m-1}$ and $w = v - \langle v, e_0 \rangle e_0 - \dots - \langle v, e_{m-1} \rangle e_{m-1}$, then we can write $v = u + w$, it is easy to show w is orthogonal to e_k . ■

Theorem 6.52 (Double Orthogonal Complement Elimination): Let U a finite subspace of V , then:

$$U = (U^\perp)^\perp$$

Proof.

- For any $u \in U$, we will show that $u \in (U^\perp)^\perp$, that is, u is orthogonal to $U^\perp$. For any $v \in U^\perp$, we know $\langle u, v \rangle = 0$ since $v \in U^\perp$, thus u is orthogonal to $U^\perp$, therefore $u \in (U^\perp)^\perp$ and $U \subseteq (U^\perp)^\perp$.
- For any $u \in (U^\perp)^\perp$, we know u is orthogonal to $U^\perp$, that means for any $v \in U^\perp$, $\langle v, u \rangle = 0$. Then by $V = U \oplus U^\perp$. We know $u \in U$, since we can write $u = ae + bf$ where $e \in U$ and $f \in U^\perp$, and we know $bf = 0$ since $\langle v, u \rangle = \langle v, ae + bf \rangle = \langle v, ae \rangle + \langle v, bf \rangle = b\langle v, f \rangle = 0$. Thus $u = ae \in U$.

Another clever proof of the second part is: Let $v \in (U^\perp)^\perp$, then $v = u + w$ where $u \in U$ and $w \in U^\perp$, then $v - u = w \in U^\perp$. We know $v \in (U^\perp)^\perp$ and $u \in U \subseteq (U^\perp)^\perp$, thus $v - u \in (U^\perp)^\perp \cap U^\perp$, then $v - u \in \{0\}$ (use $U^\perp$ as the subspace of V instead of U), thus $v = u \in U$. ■

Theorem 6.54: Let U a finite subspace of V , then

$$U^\perp = \{0\} \iff U = V$$

Proof.

$$\begin{aligned} U^\perp &= \{0\} \\ (U^\perp)^\perp &= \{0\}^\perp \quad \text{by } -^\perp \\ U &= \{0\}^\perp \quad \text{by Double Orthogonal Complement Elimination} \\ U &= V \quad \text{by property of orthogonal complement} \end{aligned}$$

■

Definition 6.55: Let U a finite subspace of V , an orthogonal projection $P_U \in \mathcal{L}(V)$ that maps V to U is defined by:

$$\begin{aligned} P_{U(v)} &= u \\ \text{where} \\ v &= u + w, u \in U, w \in U^\perp \end{aligned}$$

Theorem 6.57 (Properties of Orthogonal Projection): Let U a finite subspace of V , then

- a) $P_U \in \mathcal{L}(V)$
- b) For any $u \in U$, $P_U u = u$
- c) For any $w \in U^\perp$, $P_U w = 0$
- d) $\text{range } P_U = U$
- e) $\text{null } P_U = U^\perp$
- f) For any $v \in V$, $v - P_U v \in U^\perp$
- g) $P_U^2 = P_U$
- h) For all $v \in V$, $\|P_U v\| \leq \|v\|$
- i) Let $e_0, \dots, e_{m-1}$ a orthonormal basis of U and $v \in V$, then:

$$P_U v = \langle v, e_0 \rangle e_0 + \dots + \langle v, e_{m-1} \rangle e_{m-1}$$

Proof.

- Additivity by the unique representation, Homogeneity is trivial.
- For any $u \in U$, $u = v + w$ where $v \in U$ and $w \in U^\perp$, then $w = 0$ (cause $0 = \langle u, w \rangle = \langle v + w, w \rangle = \langle v, w \rangle + \langle w, w \rangle = 0 + \langle w, w \rangle$), therefore $u = v = P_U u$
- For any $w \in U^\perp$, $w = u + v$ where $u \in U$ and $v \in U^\perp$, then $u = 0$ since $0 = \langle w, u \rangle = \langle u + v, u \rangle = \langle u, u \rangle + \langle v, u \rangle = \langle u, u \rangle + 0 = \langle u, u \rangle$
- By definition $P_U U \subseteq U$, then by b) we know $U \subseteq P_U U$, thus $\text{range } P_U = U$.
- By c) we know $U^\perp \subseteq \text{null } P_U$, and for any $v \in V$ such that $P_U v = 0$, that means $v = u + w$ where $u = 0 \in U$ and $w \in U^\perp$, thus $v = w \in U^\perp$ and therefore $\text{null } P_U \subseteq U^\perp$.
- Trivial, by definition of P_U .
- Trivial, by b).
- Trivial, by definition of P_U .
- We can write v in form of $v = \underbrace{\langle v, e_0 \rangle e_0 + \dots + \langle v, e_{m-1} \rangle e_{m-1}}_u + \underbrace{v - \langle v, e_0 \rangle e_0 - \dots - \langle v, e_{m-1} \rangle e_{m-1}}_w$, it is easy to show that $u \in U$ and $w \in U^\perp$.

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Theorem 6.61 (Shortest Distance To Subspace): Let U a finite subspace of V , $v \in V$ and $u \in U$. Then

$$\|v - P_U v\| \leq \|v - u\|$$

The inequality is equal if and only $u = P_U v$.

Proof. This **clever** proof comes from the textbook.

$$\begin{aligned} \|v - P_U v\|^2 &\leq \|v - P_U v\|^2 + \|P_U v - u\|^2 \\ &= \|v - P_U v + P_U v - u\|^2 \\ &= \|v - u\|^2 \end{aligned}$$

The second line is by $v - P_U v \in U^\perp$ is orthogonal to $P_U v - u \in U$. And the inequality is equal if and only if $\|P_U v - u\| = 0$, which is equivalent to $u = P_U v$.

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Theorem 6.67 (Strict on $(\text{null } T)^\perp$ is bijective): Let V finite and $T \in \mathcal{L}(V, W)$. Then $T|_{(\text{null } T)^\perp} : (\text{null } T)^\perp \rightarrow \text{range } T$ is bijective.

Proof. Recall that $\dim (\text{null } T)^\perp = \dim V - \dim \text{null } T = \dim \text{range } T$. Let $v \in (\text{null } T)^\perp$ such that $T|_{(\text{null } T)^\perp} v = 0$, then $Tv = 0$, thus $v \in \text{null } T$, therefore $v \in \text{null } T \cap (\text{null } T)^\perp$, hence $v = 0$. Then $\text{null}(T|_{(\text{null } T)^\perp}) = \{0\}$, therefore $T|_{(\text{null } T)^\perp}$ is injective.

Surjective is trivial by V is finite and $T|_{(\text{null } T)^\perp}$ is injective. ■

Definition 6.68 (Pseudoinverse): Let V finite, $T \in \mathcal{L}(V, W)$. The pseudoinverse of T is defined by:

$$T^\dagger w = \left(T|_{(\text{null } T)^\perp}\right)^{-1} P_{\text{range } T} w$$

Informally, $T(T^\dagger w)$ is the nearest element to w (by Shortest Distance To Subspace)

Theorem 6.69 (Properties of Pseudoinverse): Let V finite and $T \in \mathcal{L}(V, W)$.

- a) If T is invertible, then $T^\dagger = T^{-1}$
- b) $TT^\dagger = P_{\text{range } T}$ = orthogonal projection from W to $\text{range } T$
- c) $T^\dagger T = P_{(\text{null } T)^\perp}$ = orthogonal projection from V to $(\text{null } T)^\perp$.

Proof.

- 1. If T is invertible, then T is surjective, thus $P_{\text{range } T} w = w$ and $T|_{(\text{null } T)^\perp} = T$ since $\text{null } T = \{0\}$ implies $(\text{null } T)^\perp = V$. Then $T^\dagger = T^{-1} \circ I$.
- 2. Trivial
- 3. By unfolding, we get $T^\dagger T = \left(T|_{(\text{null } T)^\perp}\right)^{-1} \circ T$. We want to show that $T(P_{(\text{null } T)^\perp} v) = Tv$, then we know $\left(T|_{(\text{null } T)^\perp}\right)^{-1}(Tv) = P_{(\text{null } T)^\perp} v$. We can see

$$\begin{aligned} T(P_{(\text{null } T)^\perp} v) &= T(v - (v - P_{(\text{null } T)^\perp} v)) \\ &= Tv - T(v - P_{(\text{null } T)^\perp} v) \\ &= Tv \end{aligned}$$

the second equation is by Properties of Orthogonal Projection (f). ■

Theorem 6.70 (Pseudoinverse is the best solution): Let V finite, $T \in \mathcal{L}(V, W)$, $w \in W$.

a) Let $v \in V$, then

$$\|T(T^\dagger w) - w\| \leq \|Tv - w\|$$

the inequality is equal if and only if $v \in T^\dagger w + \text{null } T$.

b) Let $v \in T^\dagger w + \text{null } T$, then

$$\|T^\dagger w\| \leq \|v\|$$

the inequality is equal if and only if $v = T^\dagger w$.

Note that $T^\dagger w + \text{null } T$ is a translate, not the sum of two subspaces.

Proof.

$$\begin{aligned} 1. \quad \|T(T^\dagger w) - w\| &= \|P_{\text{range } T} w - w\| \\ &= \|w - P_{\text{range } T} w\| \\ &\leq \|w - Tv\| \quad \text{by Shortest Distance To Subspace} \\ &= \|Tv - w\| \end{aligned}$$

and the inequality equal if and only if $Tv = P_{\text{range } T} w$. By $w' = (T|_{(\text{null } T)^\perp})^{-1} P_{\text{range } T} w = T^\dagger w$ we know $v = w' + u$ where $w' \in (\text{null } T)^\perp$ and $u \in \text{null } T$. It is easy to show that w' is unique such that $v = w' + u$ by Strict on $(\text{null } T)^\perp$ is bijective. Thus $v \in T^\dagger w + \text{null } T$ if and only if $Tv = P_{\text{range } T} w$

2. Let $u \in \text{null } T$ such that $v = T^\dagger w + u$

$$\|v\|^2 = \|T^\dagger w + u\|^2 = \|T^\dagger w\|^2 + \|u\|^2$$

thus $\|T^\dagger w\| \leq \|v\|$, equal if and only if $u = 0$, which is equivalent to $v = T^\dagger w$.

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